Prerequisites & Hardware

- **Hardware:** ESP32 Development Board, USB Data Cable (ensure it transfers data, not just power).
- **Software:** Arduino IDE installed on your computer.
- **Network:** A 2.4 GHz Wi-Fi network (ESP32 does not support 5 GHz Wi-Fi).

Step 1: Configure the Arduino IDE

1. **Install ESP32 Board Support:**
    - Open Arduino IDE. Go to **File > Preferences**.
    - Paste this URL into *Additional Boards Manager URLs*: `[https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json](https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json)`
    - Go to **Tools > Board > Boards Manager**, search for **esp32** by Espressif, and click **Install**.
    - Connect your ESP32 to your computer via USB. Select your board under **Tools > Board > esp32 > ESP32 Dev Module**, and choose your port under **Tools > Port**.
2. **Install the MQTT Library:**
    - Go to **Tools > Manage Libraries...** (or press `Ctrl+Shift+I`).
    - Search for **PubSubClient** (by Nick O'Leary).
    - Click **Install**.

Step 2: Upload the ESP32 Code

Replace `"YOUR_WIFI_SSID"` and `"YOUR_WIFI_PASSWORD"` with your network credentials, then click the **Upload** arrow button in the Arduino IDE.

[Check the MQTT.ino file for the code]

Step 3: Open the Web Test Dashboard

1. In your web browser, open the HiveMQ Web Client.
2. Keep the default settings (**Host:** `mqtt-dashboard.com`, **Port:** `8884`).
3. Click the blue **Connect** button. The red indicator dot in the top-right will turn green and say **connected**.

Step 4: Run the Trial and Verify

1. **Open Serial Monitor:** In Arduino IDE, click **Tools > Serial Monitor** and set the baud rate dropdown in the top-right to **115200 baud**. You will see `WiFi connected!` followed by `Attempting MQTT connection...connected!`.
2. **Receive ESP32 Data on Web:**
    - On the HiveMQ web page, expand the **Subscriptions** panel.
    - Click **Add New Topic Subscription**.
    - Type `esp32/trial/out` into the Topic field and click **Subscribe**.
    - Every 5 seconds, `"Hello from ESP32!"` will pop up in the **Messages** panel.
3. **Send Web Data to ESP32:**
    - On the HiveMQ web page, expand the **Publish** panel.
    - Enter `esp32/trial/in` into the Topic field.
    - Enter any message in the Message box (e.g., `Hello from browser!`).
    - Click **Publish**.
    - Look at your Arduino Serial Monitor—you will immediately see `Message arrived on topic [esp32/trial/in]: Hello from browser!`.